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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

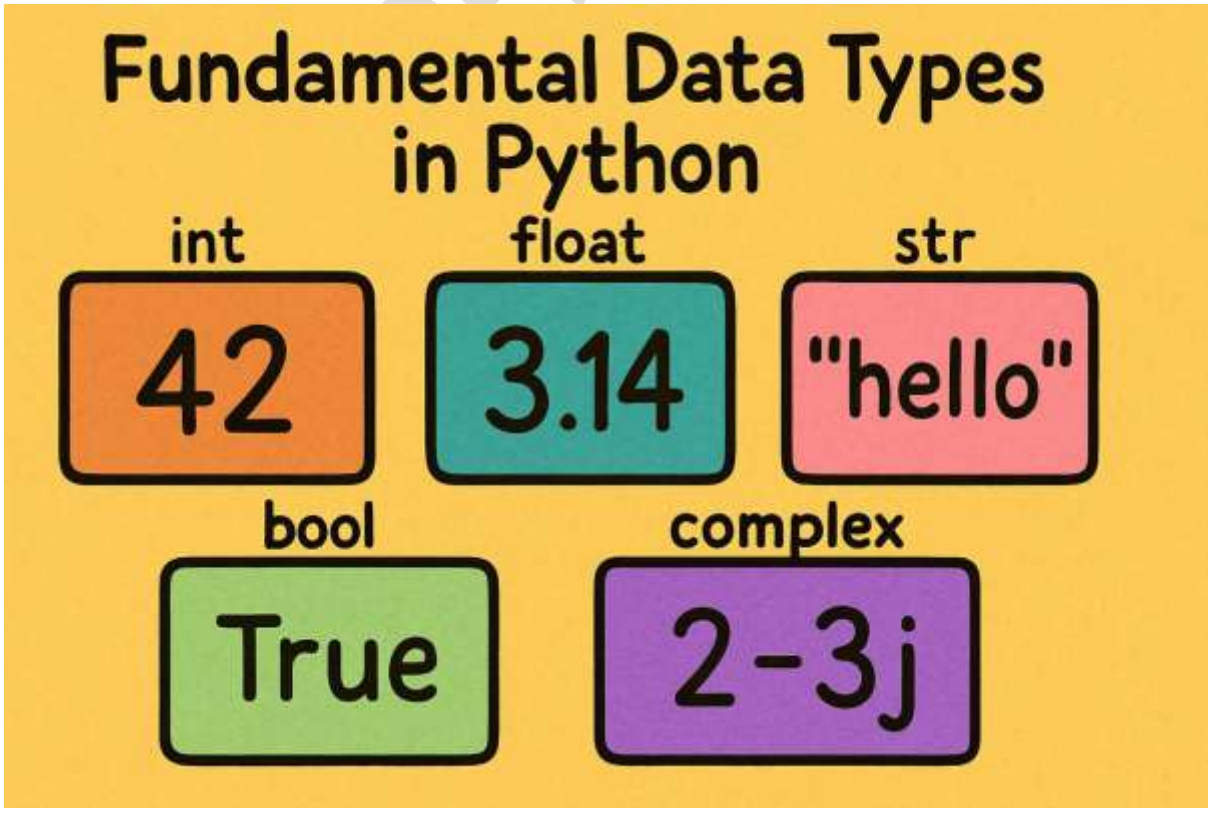
Classes : Offline & Online

Daily Teaching : 2 Hours

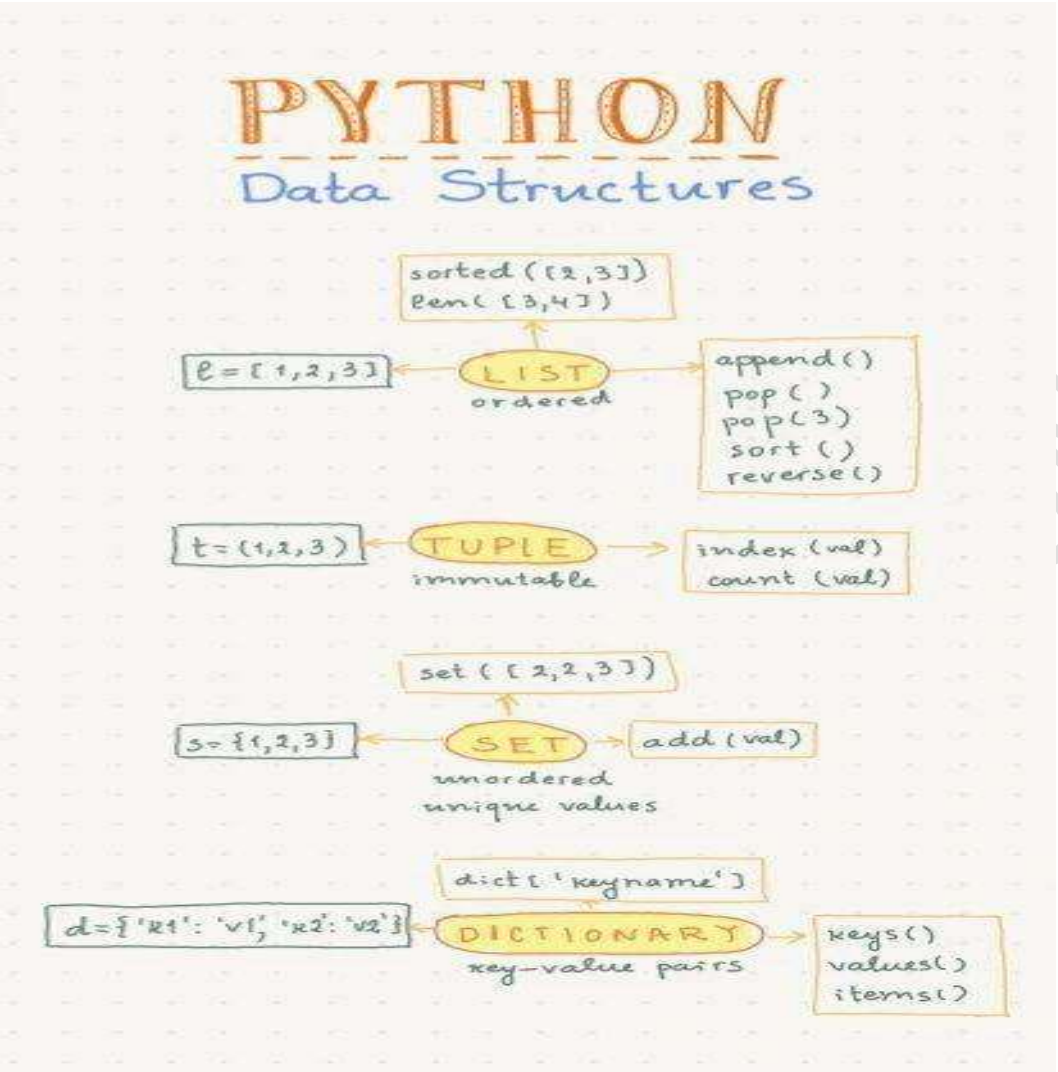
Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 16: Data Types (Python)



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PYTHON
Data Structures
sorted([2,3])
len([3,4])
l=[1,2,3]
LIST
ordered
append()
pop()
pop(3)
sort()
reverse()
t=(1,2,3)
TUPLE
immutable
index(val)
count(val)
set([2,2,3])
s={1,2,3}
SET
unordered
unique values
add(val)
dict['keyname']
d={'k1': 'v1', 'k2': 'v2'}
DICTIONARY
key-value pairs
keys()
values()
items()



Learning Objectives

By the end of this lesson, students will be able to:

- Understand what a data type is.
- Identify different Python data types.
- Store different kinds of values in variables.
- Use numbers, text, and True/False values.
- Understand lists, tuples, dictionaries, and sets.
- Write simple Python programs using different data types.

What is a Data Type?

A **Data Type** tells Python **what kind of information** is stored in a variable.

Think of a school bag:

🎒 Different pockets store different things.

- Books 📚
- Water Bottle 🥤
- Lunch Box 🍱

Similarly, Python stores different types of data.

Example

```
name = "Rahul"

age = 13

height = 4.9

is_student = True
```

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Python automatically understands the type of each value.

Why are Data Types Important?

They help Python know:

- How to store data
- How much memory to use
- What operations are allowed

Example

You can add numbers:

10 + 20

But you cannot add a number and text directly:

10 + "20"

This gives an error.

Python Basic Data Types

Data Type	Example	Meaning
int	25	Whole Number
float	3.14	Decimal Number
str	"Hello"	Text
bool	True	True or False
list	[1,2,3]	Collection
tuple	(1,2,3)	Fixed Collection

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Data Type	Example	Meaning
dict	{"name":"Ali"}	Key-Value Pair
set	{1,2,3}	Unique Values

1 Integer (int)

Integers are whole numbers.

Examples

10

50

-25

1000

Program

```
age = 13


print(age)
print(type(age))
```

Output

```
13
<class 'int'>
```

2 Float (float)

Float means decimal numbers.

Examples

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2.5

9.99

45.7

Program

height = 5.2

print(height)

print(type(height))

Output

5.2

<class 'float'>

3️⃣ String (str)

A String is text.

Always written inside quotes.

Examples

"Python"

"Hello"

"School"

Program

name = "Anjali"

print(name)

print(type(name))

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Output

Anjali

<class 'str'>

4️⃣ Boolean (bool)

Boolean has only two values.

- True
- False

Program

```
is_student = True


print(is_student)

print(type(is_student))
```

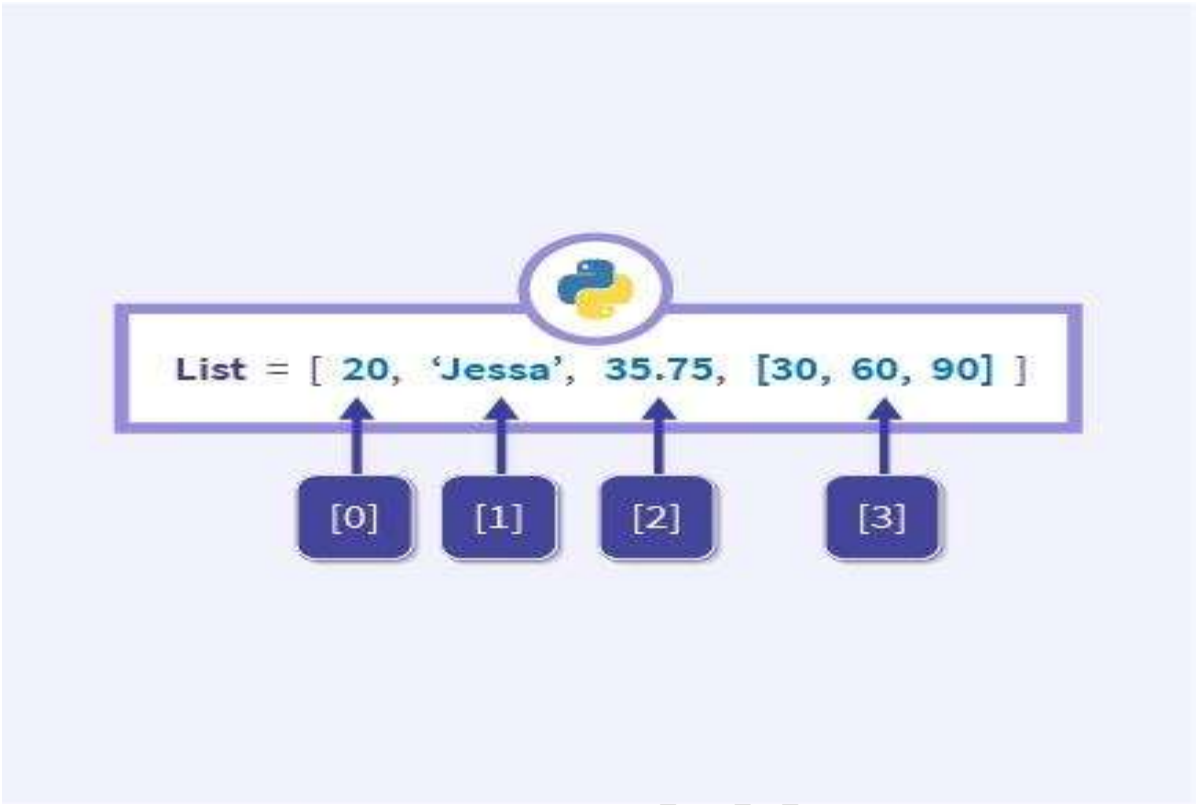
Output

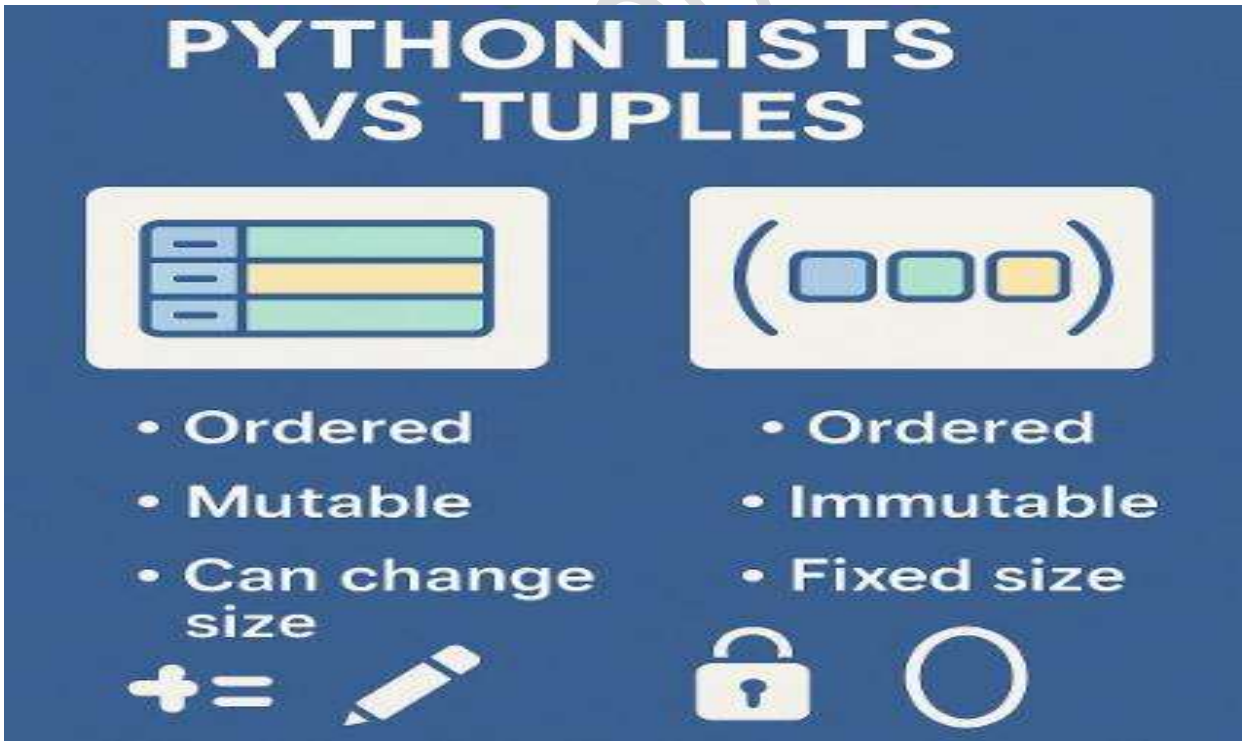
True

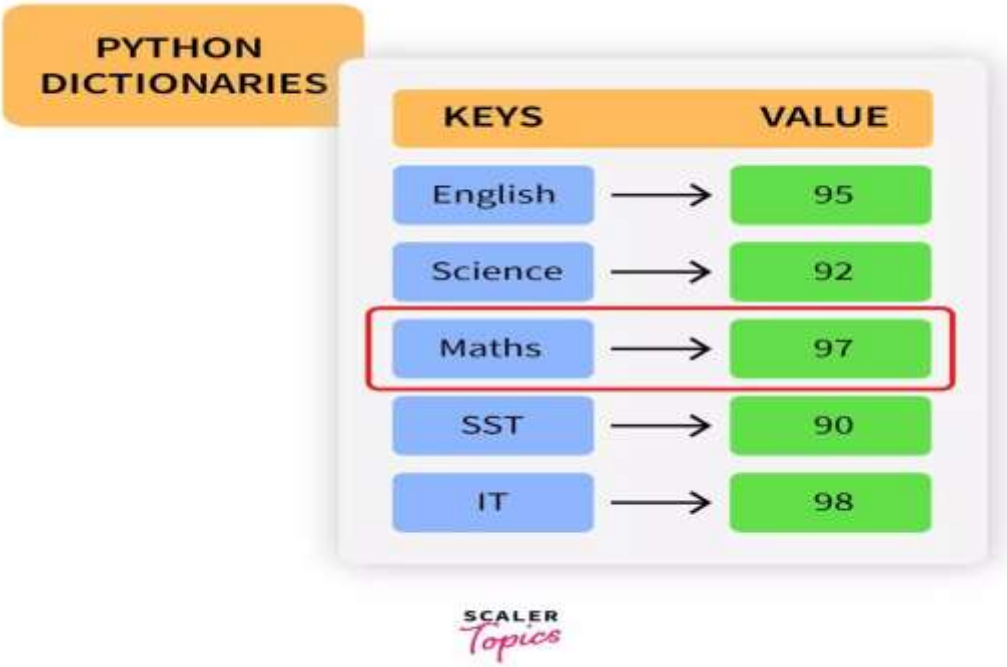
<class 'bool'>

Collections in Python

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List = [20, 'Jessa', 35.75, [30, 60, 90]]
[0]
[1]
[2]
[3]


PYTHON LISTS
VS TUPLES
• Ordered
• Mutable
• Can change size
• Ordered
• Immutable
• Fixed size



Collections store multiple values together.

5 List

A List stores many items.

Uses square brackets.

fruits = ["Apple", "Banana", "Mango"]

print(fruits)

Output

['Apple', 'Banana', 'Mango']

Lists can be changed.

Access List Items

fruits = ["Apple","Banana","Mango"]

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```
print(fruits[0])
```

Output

Apple

6 Tuple

Tuple also stores multiple items.

Uses parentheses.

```
numbers = (10,20,30)


print(numbers)
```

Output

(10,20,30)

Tuples cannot be changed after creation.

7 Dictionary

Dictionary stores data in **Key : Value** form.

```
student = {
   "Name":"Rahul",
   "Age":13,
   "Class":7
}


print(student)
```

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Output

{'Name':'Rahul','Age':13,'Class':7}

Access value

print(student["Name"])

Output

Rahul

8 Set

Set stores unique values.

Duplicates are removed.

numbers = {1,2,3,3,4,5}

print(numbers)

Output

{1,2,3,4,5}

Checking Data Type

Use **type()**

a = 25

print(type(a))

Output

<class 'int'>

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Examples

```
print(type(10))


print(type(5.5))


print(type("Python"))


print(type(True))
```

Output

```
<class 'int'>
<class 'float'>
<class 'str'>
<class 'bool'>
```

Activity 1

Guess the Data Type

Value	Answer
25	Integer
4.5	Float
"India"	String
True	Boolean
[1,2,3]	List
(1,2,3)	Tuple

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Value	Answer
{"A":1}	Dictionary
{1,2,3}	Set

Activity 2

Write a program.

```
name = "Rahul"

age = 13

height = 5.1

is_student = True

print(name)

print(age)

print(height)

print(is_student)
```

Activity 3

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Find the Data Type

```
a = 10


b = 3.14


c = "Python"


d = False


print(type(a))


print(type(b))


print(type(c))


print(type(d))
```

Practice Exercise

Create the following variables.

student_name

school_name

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marks

percentage

is_present

Print all values and their types.

Mini Project

Student Information Card

name = "Aisha"

age = 13

marks = 95.5

is_student = True

subjects = ["Math","Science","English"]

print("Student Details")

print(name)

print(age)

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```
print(marks)

print(is_student)

print(subjects)
```

Output

```
Student Details

Aisha

13

95.5

True

['Math','Science','English']
```

Real-Life Examples

Real Life	**Python Data Type**
Student Name	String
Roll Number	Integer

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Real Life	Python Data Type
Height	Float
Attendance	Boolean
Subjects	List
School Details	Dictionary
Unique Club IDs	Set

Common Mistakes

❌ Forgetting quotation marks

```
name = Rahul
```

✓ Correct

```
name = "Rahul"
```

❌ Mixing number and text

```
10 + "20"
```

✓ Convert first

```
10 + int("20")
```

Output

```
30
```

Classroom Activity

Create variables for:

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- Your Name
- Age
- Favourite Subject
- Favourite Game
- Marks
- Height
- Present (True/False)

Print them with type().

Quick Quiz

Choose the Correct Answer

1. Which data type stores text?

A) int

B) float

C) string ✅

D) bool

2. Which stores True or False?

A) bool ✅

B) int

C) float

D) list

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3. Which brackets are used for List?

A) ()

B) [] ✅

C) {}

D) <>

4. Which data type stores key-value pairs?

A) List

B) Dictionary ✅

C) Float

D) String

5. Which function checks the data type?

A) print()

B) input()

C) type() ✅

D) len()

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Summary

✅ Data Types tell Python what kind of data is stored.

✅ Main Python data types:

- Integer (int)
- Float (float)

- String (str)
- Boolean (bool)
- List (list)
- Tuple (tuple)
- Dictionary (dict)
- Set (set)

✅ Use type() to check a variable's data type.

✅ Choosing the correct data type makes programs easier to understand and more efficient.

Home Assignment

1. Create variables for your personal details (name, age, height, and whether you like Python) and print their values and data types.
2. Make a list of your five favorite fruits and print the second fruit.
3. Create a dictionary containing your **Name**, **Class**, **School**, and **City**, then print each value separately.
4. Create a set with duplicate numbers and observe how Python removes repeated values.
5. Write a short program that uses at least one example of each basic Python data type covered in this lesson.

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